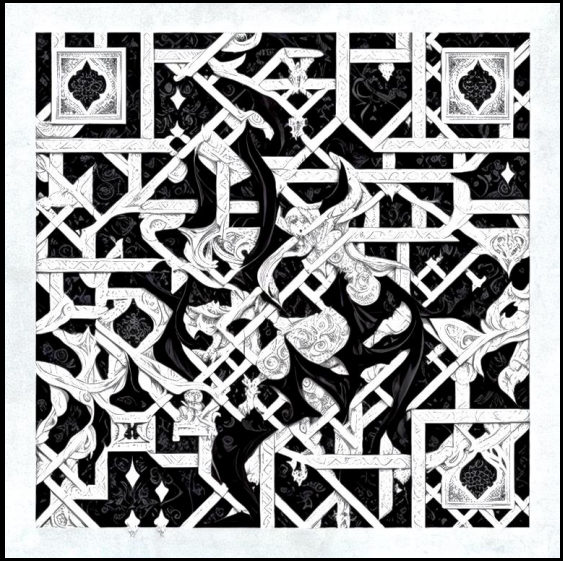


Embellir des QR Codes avec la GenAI

Raphaël Semeteys
COTRECS
14 mars 2025

QR Codes

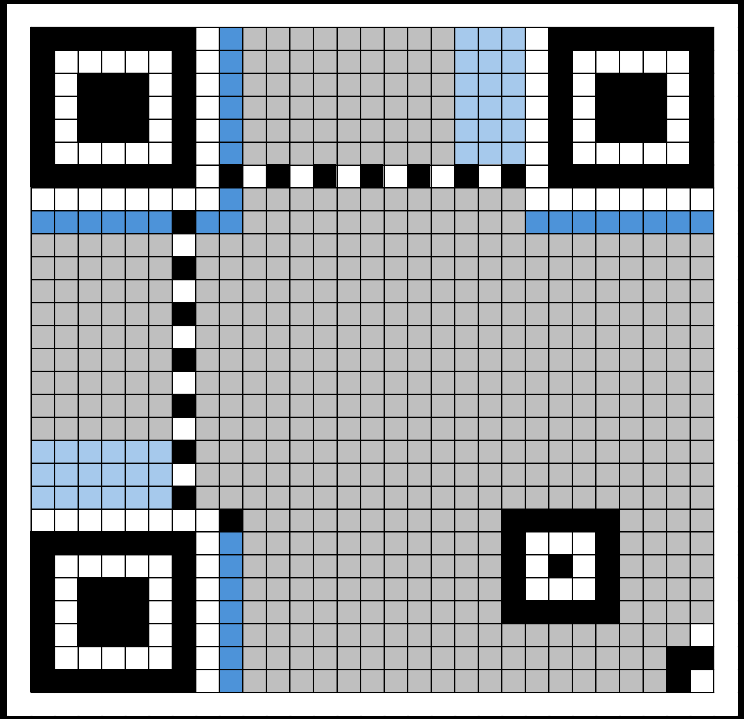
Quick Response Codes

- **Histoire**
  - Crée en 1994 dans une filiale de Toyota
  - Standards ISO/IEC 18004:2000/2006/2015/2024
  - Inspiré du jeu Go
- **Évolutionn 2D des code-barres**
  - Motif de carrés noirs sur fond blanc
  - Contenu numérique, alphanumérique, binaire ou Kanji
- **Différentes tailles (ou versions)**
  - De v1 (21x21) à v40 (177x177)
  - Chaque version est plus large de 4 que la précédente
- **Mécanisme intégré de correction d'erreurs**



Anatomie d’un QR Code

- **Lisibilité par les scanneurs**
  - Identification et focus
    - *Quiet Zone*
    - 3 *Positional Markers* (les « yeux »)
  - *Alignment patterns* pour gérer l’orientation
- **Modules**
  - Le contenu est stocké au format binaire
  - Carré noir = 1 | Carré blanc = 0
  - *Timing pattern* pour préciser la taille d’un module
- **Format** : *Level of Error Correction* et *Mask pattern*
- **Version**



v3: 29x29

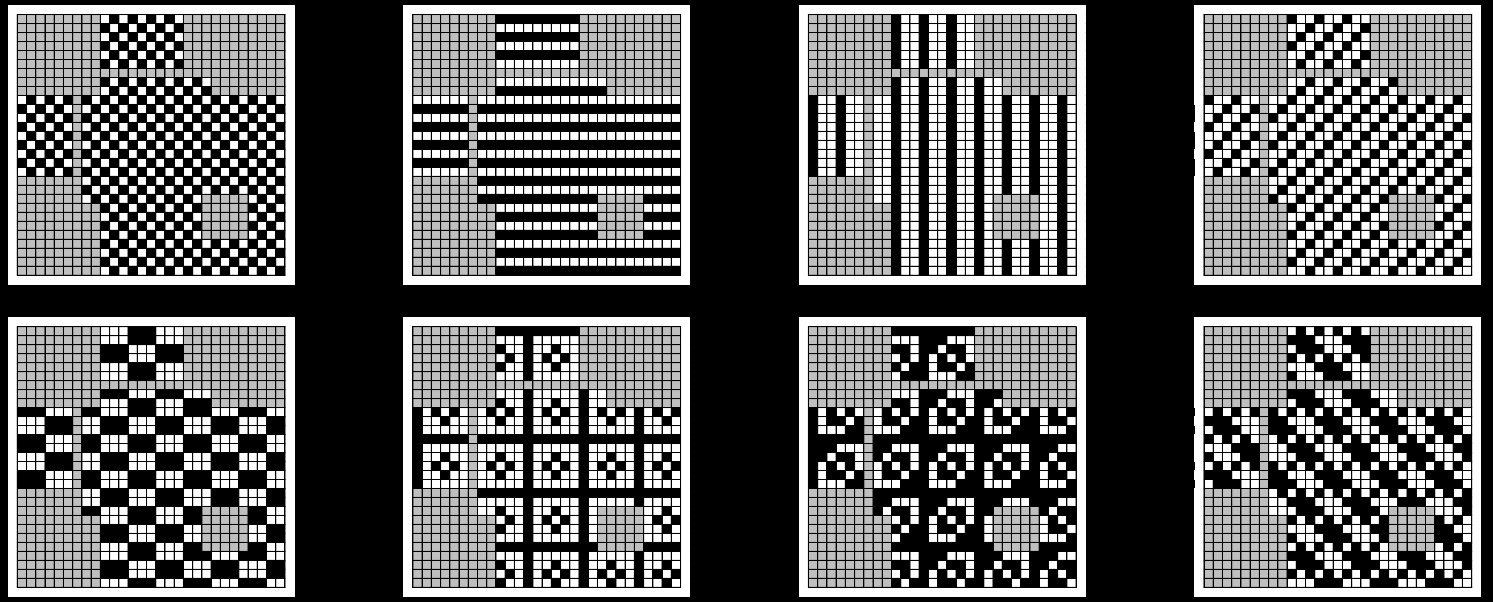
Correction d’erreur

- **Des données supplémentaires sont ajoutées**
  - Sous forme de nouvelles lignes et colonnes
  - Cela augmente la taille du QR Code
  - Compense d’éventuelles données manquantes à la lecture
- **Mécanisme mis en œuvre**
  - *Error Correction Codes* de Reed-Solomon
    - Maths à base de polynômes et de corps finis
    - Utilisé par stockage (CD/DVD, RAID6) et réseau (DSL, sat)
  - *Codewords* de 8 bits pour données et codes de correction
- **4 niveaux de correction possibles**
- **Utilisé pour injecter des images** (vues comme des erreurs)



Niveau	% erreur
Low	~7%
Medium	~15%
Quartile	~25%
High	~30%

Mask Patterns



- **Masques d’optimisation appliqués sur les données (XOR)**
  - 8 possibilités
  - Permet de sélectionner le masque le plus efficace

Contenu

- **Capacité de stockage** varie en fonction de
  - Version (taille)
  - Type of données
  - Niveau de correction

- **QR Codes dynamiques**
  - Stockent des liens courts
  - Redirection vers le contenu final
    - Plus court et efficace
    - Contenu « modifiable »
    - *Tracking* et *Analytics*
    - Protection par mot de passe, date d’expriration

Version	Error Correction Level	Numeric Mode	Alphanumeric Mode	Byte Mode	Kanji Mode
1	L	41	25	17	10
	M	34	20	14	8
	Q	27	16	11	7
	H	17	10	7	4
2	L	77	47	32	20
	M	63	38	26	16
	Q	48	29	20	12
	H	34	20	14	8
3	L	127	77	53	32
	M	101	61	42	26
	Q	77	47	32	20
	H	58	35	24	15
4	L	187	114	78	48
	M	149	90	62	38
	Q	111	67	46	28
	H	82	50	34	21
5	L	255	154	106	65
	M	202	122	84	52
	Q	144	87	60	37
	H	106	64	44	27
6	L	322	195	134	82
	M	255	154	106	65
	Q	178	108	74	45
	H	139	84	58	36
	L	370		154	

35	Q	3081			790
	H	2361	1431	983	605
36	L	5836	3537	2431	1496
	M	4588	2780	1911	1176
	Q	3244	1966	1351	832
	H	2524	1530	1051	647
37	L	6153	3729	2563	1577
	M	4775	2894	1989	1224
	Q	3417	2071	1423	876
	H	2625	1591	1093	673
38	L	6479	3927	2699	1661
	M	5039	3054	2099	1292
	Q	3599	2181	1499	923
	H	2735	1658	1139	701
39	L	6743	4087	2809	1729
	M	5313	3220	2213	1362
	Q	3791	2298	1579	972
	H	2927	1774	1219	750
40	L	7089	4296	2953	1817
	M	5596	3391	2331	1435
	Q	3993	2420	1663	1024
	H	3057	1852	1273	784

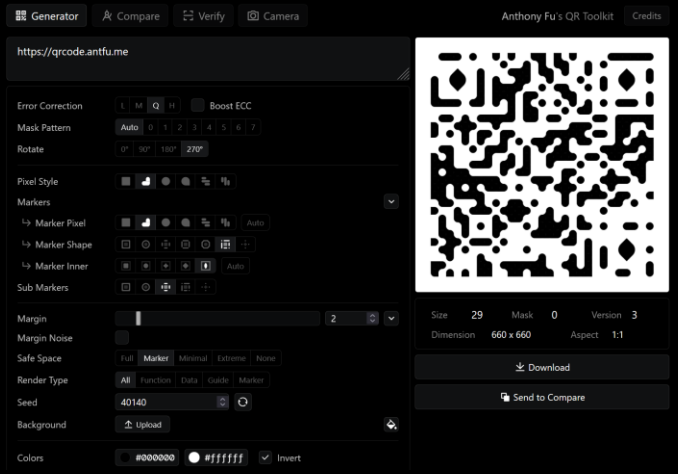
Générer des QR Codes

- **De nombreuses implémentations sont disponibles**

- **QR-Code-generator** : ma librairie préférée
  - Supporte 40 versions, 4 niveaux de correction, le *masking*
  - Tous les types de donnée
  - Java, TypeScript/JavaScript, Python, Rust, C++, C
  - Licence MIT
- **QR Toolkit** : mon application préférée
  - Supporte toutes les versions et la plupart des fonctionnalités
  - Personnalisation des *markers* et des modules
  - Génération, vérification et comparaison de QR Codes
  - Application Vue/Nuxt sous licence MIT



nayuki.io/page/qr-code-generator-library



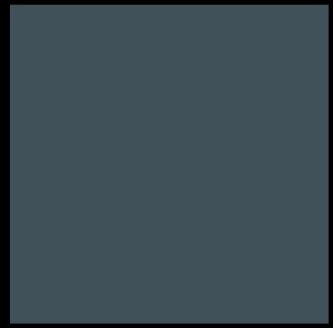
qrcode.antfu.me

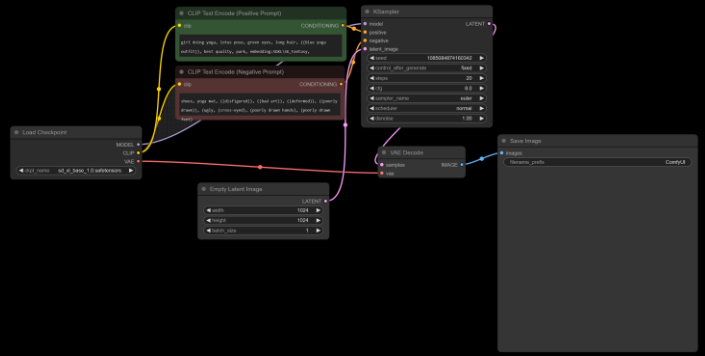
Generative AI

Modifier un code QR existant avec la GenAI tout en restant lisible par les scanners

Génération d’image

- **Stable Diffusion**
  - Débruitage progressif d’une image latente
  - Guidé par un prompt (T2I) et des éléments visuels (I2I)
  - Modèles « ouverts » et communautés dynamiques
- **ComfyUI**
  - IHM modulaire pour *workflows* de génération
  - Exécution locale (Windows, Mac, Linux)
  - Licence GPL 3



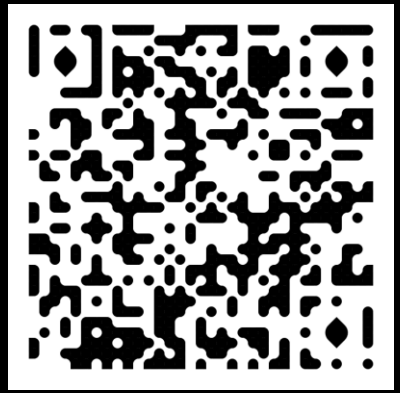


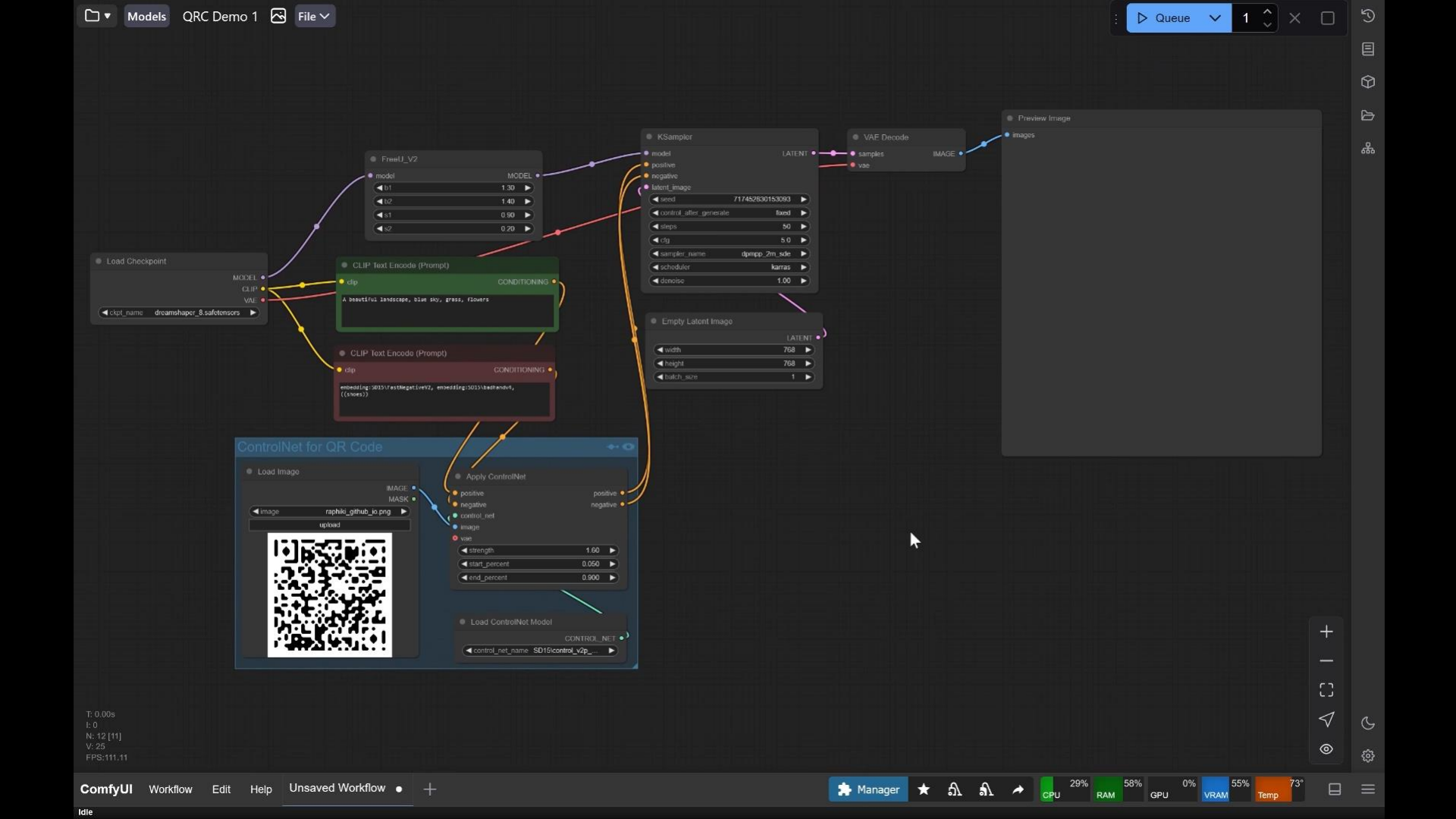
ControlNets

- **Réseaux de neurones spécialisés**
  - Guidage et contrôle complémentaires au modèle de base
  - Injectent/transfèrent certaines caractéristiques d'une image

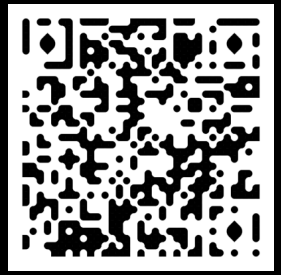
  → Approche hybride avec références textuelles et visuelles

- **ControlNet QR Code Monster**
  - "*This model is made to generate creative QR codes that still scan*"
  - Essayons-le !
  - Utilisons QR Toolkit pour générer le QR Code de base




Models
QRC Demo 1
File
Queue
1
FreeU_V2
model
MODEL
b1 1.30
b2 1.40
s1 0.90
s2 0.20
KSampler
model
positive
negative
latent_image
LATENT
seed 717452830153093
control_after_generate fixed
steps 50
cfg 5.0
sampler_name dpmpp_2m_sde
scheduler karras
denoise 1.00
VAE Decode
samples
vae
IMAGE
Preview Image
images
Load Checkpoint
MODEL
CLIP
VAE
ckpt_name dreamshaper_8.safetensors
CLIP Text Encode (Prompt)
clip
CONDITIONING
A beautiful landscape, blue sky, grass, flowers
CLIP Text Encode (Prompt)
clip
CONDITIONING
embedding:SD15\FastNegativeV2, embedding:SD15\badhandv4, ((snoes))
Empty Latent Image
LATENT
width 768
height 768
batch_size 1
ControlNet for QR Code
Load Image
IMAGE
MASK
image raphiki_github_io.png
upload
Apply ControlNet
positive
negative
control_net
image
vae
strength 1.60
start_percent 0.050
end_percent 0.900
Load ControlNet Model
CONTROL_NET
control_net_name SD15\control_v2p_...
T: 0.00s
I: 0
N: 12 [11]
V: 25
FPS:111.11
ComfyUI
Workflow
Edit
Help
Unsaved Workflow
Manager
CPU 29%
RAM 58%
GPU 0%
VRAM 55%
Temp 73°
Idle

Modifier un QR Code avec un prompt

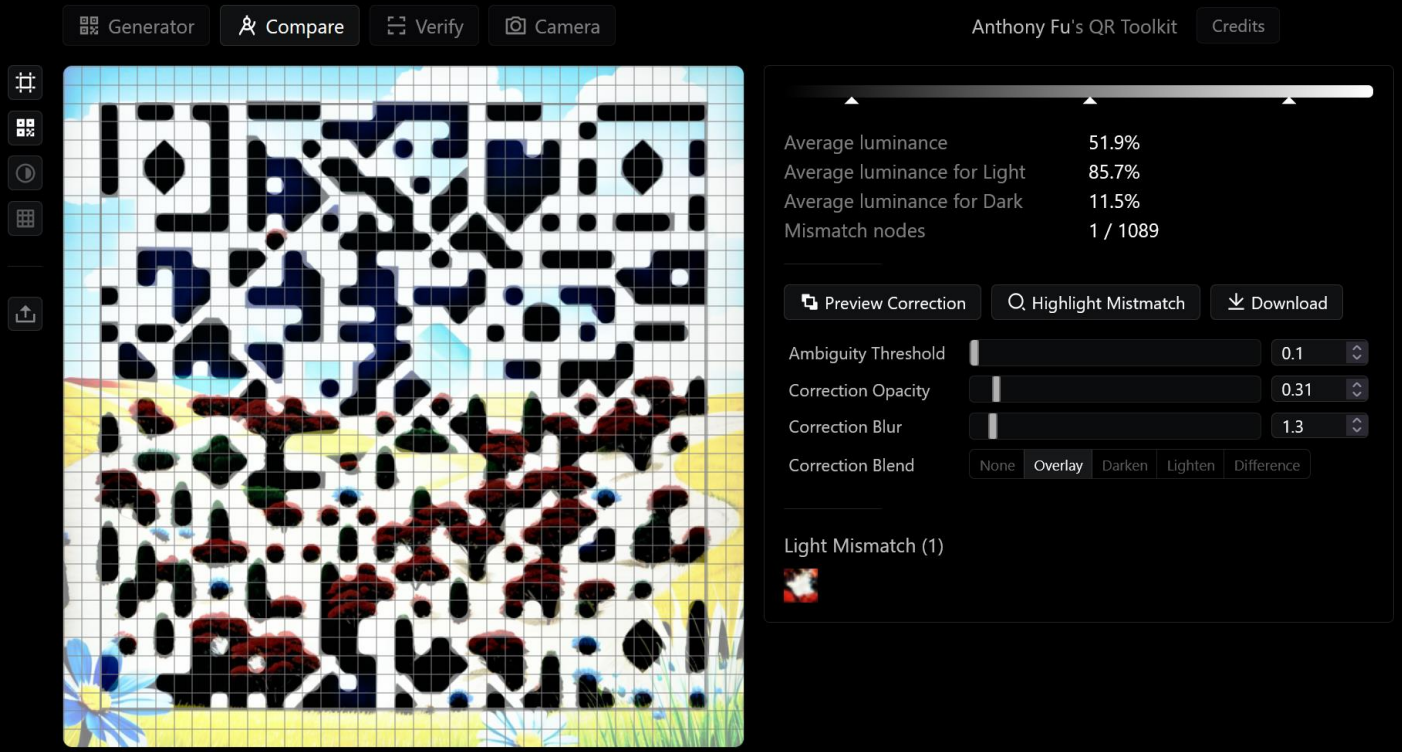


+

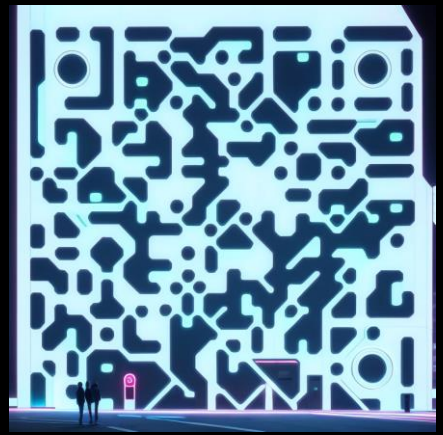
A beautiful landscape, blue sky, grass, flowers



Modifier un QR Code avec un prompt

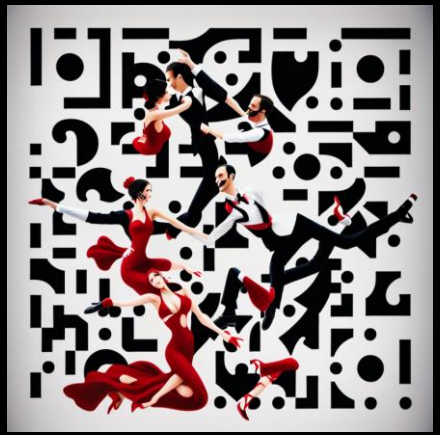


Modifier un QR Code avec un prompt



A futuristic building with neon glows

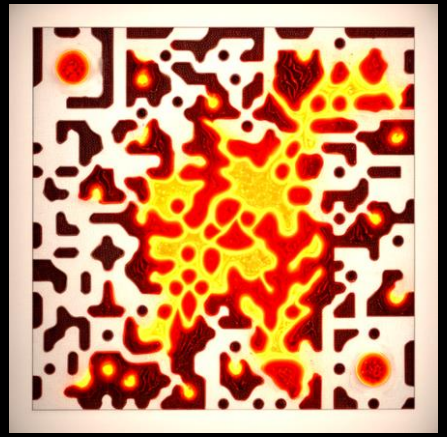
→ **Travailler sur le style**



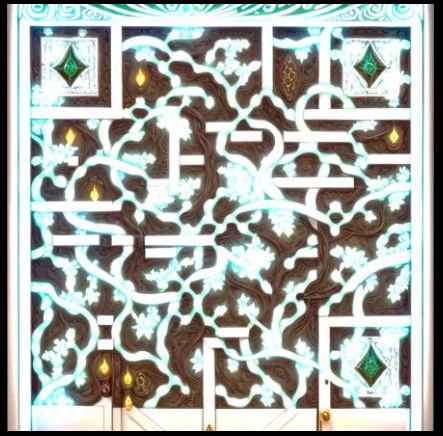
A couple dancing tango, woman with red dress, man in black tuxedo

→ **Travailler sur le contenu**

Modifier un QR Code avec un prompt venant d’un LLM



A pattern forged from molten lava, glowing with an intense fiery orange and red hue. Cracks in the surface reveal a volcanic heat, with small embers rising around it"



An elegant, glowing elven door adorned with intricate, nature-inspired patterns and shimmering silver runes. Delicate vines and luminescent flowers intertwine with the carvings, pulsating with soft emerald and sapphire light. The archway, crafted from ethereal white stone, radiates a mystical aura, with faint golden mist swirling at its base, hinting at an ancient portal to a hidden realm

Changer le style du QR Code via le prompt

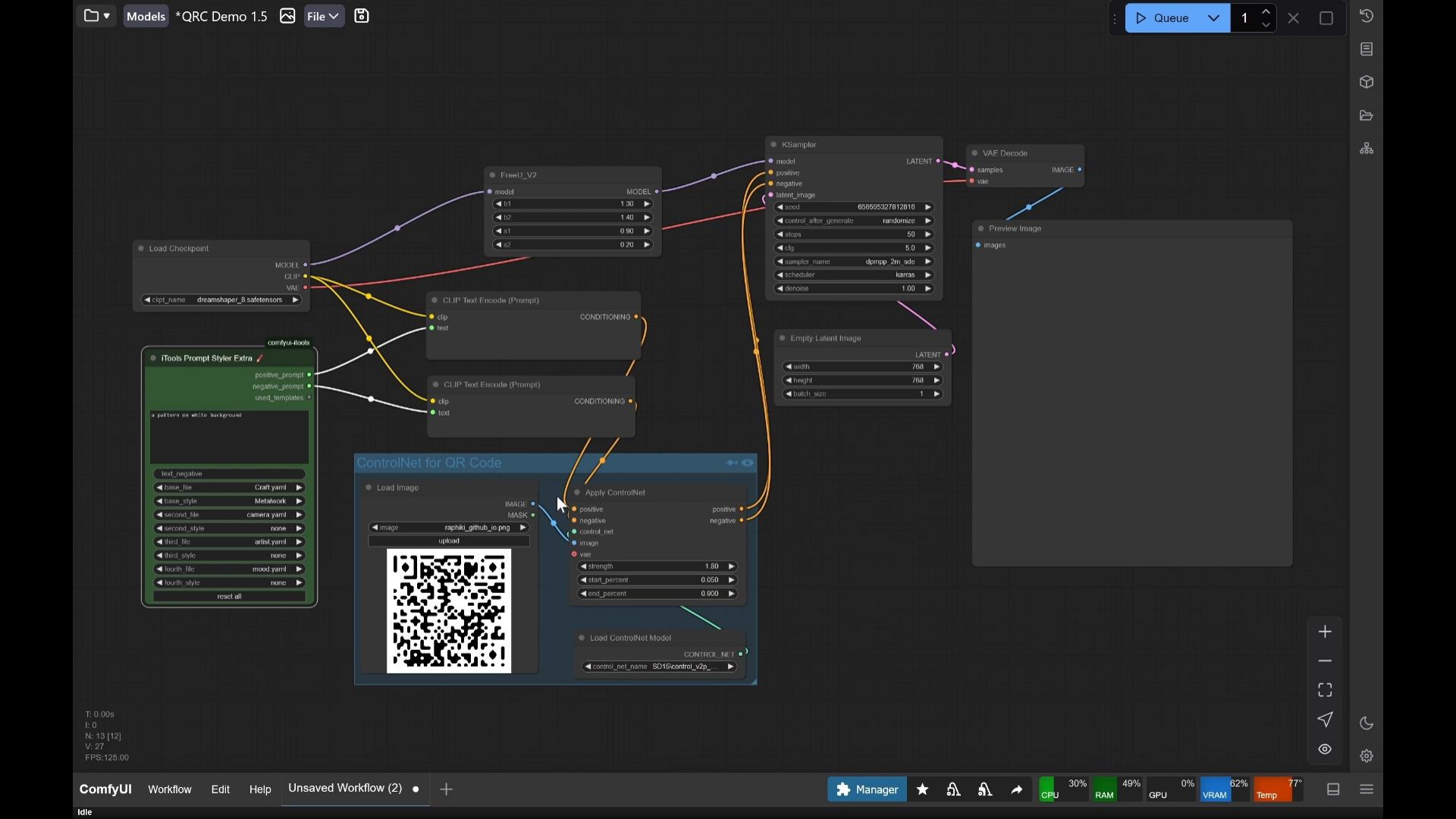



```yaml
- name: Macrame
- name: Marquetry
  negative_prompt: signature, watermark, ugly, off-center, deformed, cropped, worst quality, low quality, lowres, JPEG artifacts
  prompt: Marquetry art style of {prompt}, intricate wood inlay technique, creates decorative patterns and images using veneers of different woods, showcases craftsmanship and attention to detail

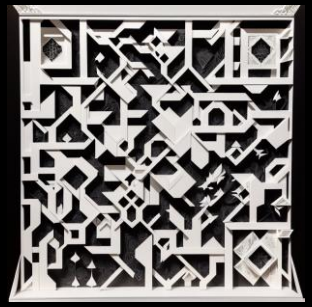
- name: Metalwork
  negative_prompt: signature, watermark, ugly, off-center, deformed, cropped, worst quality, low quality, lowres, JPEG artifacts
  prompt: Metalwork art style of {prompt}, manipulation of metal to create sculptures, showcases craftsmanship and technical skill, intricate and durable designs

- name: Mosaic
  negative_prompt: worst quality, blurry, normal quality, signature, watermark, ugly, off-center, deformed, cropped, worst quality, low quality, lowres, JPEG artifacts, low details, bad photo, bad photography, bad art
  prompt: mosaic art style of {prompt}, versatile and decorative, uses small pieces of tile, adds color and texture, glossy, sharp
```

- ***Custom Node* dans ComfyUI**
  - Prompts réutilisables pour orienter des styles
  - Nombreuses catégories : 3D, Art, Craft, Design, Drawing, Illustration, Painting, Sculpture, Vector...


Models
*QRC Demo 1.5
File
Queue
1
KSampler
model
positive
negative
latent_image
LATENT
seed 656595327812816
control_after_generate randomize
steps 50
cfg 5.0
sampler_name dpmpp_2m_sde
scheduler karras
denoise 1.00
VAE Decode
samples
vae
IMAGE
Preview Image
images
FreeU_V2
model
MODEL
b1 1.30
b2 1.40
s1 0.90
s2 0.20
Load Checkpoint
MODEL
CLIP
VAE
ckpt_name dreamshaper_8.safetensors
CLIP Text Encode (Prompt)
clip
text
CONDITIONING
Empty Latent Image
LATENT
width 768
height 768
batch_size 1
comfyui-itools
iTools Prompt Styler Extra
positive_prompt
negative_prompt
used_templates
a pattern on white background
text_negative
base_file Craft.yaml
base_style Metalwork
second_file camera.yaml
second_style none
third_file artist.yaml
third_style none
fourth_file mood.yaml
fourth_style none
reset all
ControlNet for QR Code
Load Image
IMAGE
MASK
image raphiki_github_io.png
upload
Apply ControlNet
positive
negative
control_net
image
vae
strength 1.80
start_percent 0.050
end_percent 0.900
Load ControlNet Model
CONTROL_NET
control_net_name SD15\control_v2p_...
T: 0.00s
I: 0
N: 13 [12]
V: 27
FPS:125.00
ComfyUI
Workflow
Edit
Help
Unsaved Workflow (2)
Manager
CPU 30%
RAM 49%
GPU 0%
VRAM 62%
Temp 77°
Idle

Changer le style du QR Code via le prompt



Craft | Shadowbox > Paper



Craft | Macrame



3D | Abstract > Fluid Dynamics



Art | African Traditional



Art | Rococo



Experimental. | Quantum Sketching



Basic | Expressionist



Basic | Gothic

COTRECS

WORLDLINE

Générer une image à l’intérieur du QR Code

- **ControlNets supplémentaires**
  - Pour transférer la posture : profondeur et détection de contours
  - Le ControlNet Openpose n’est pas assez précis

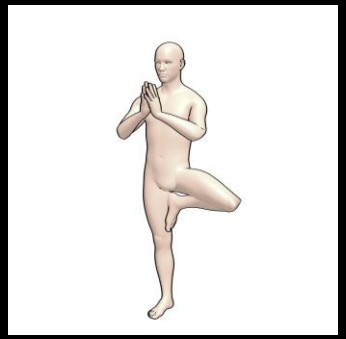


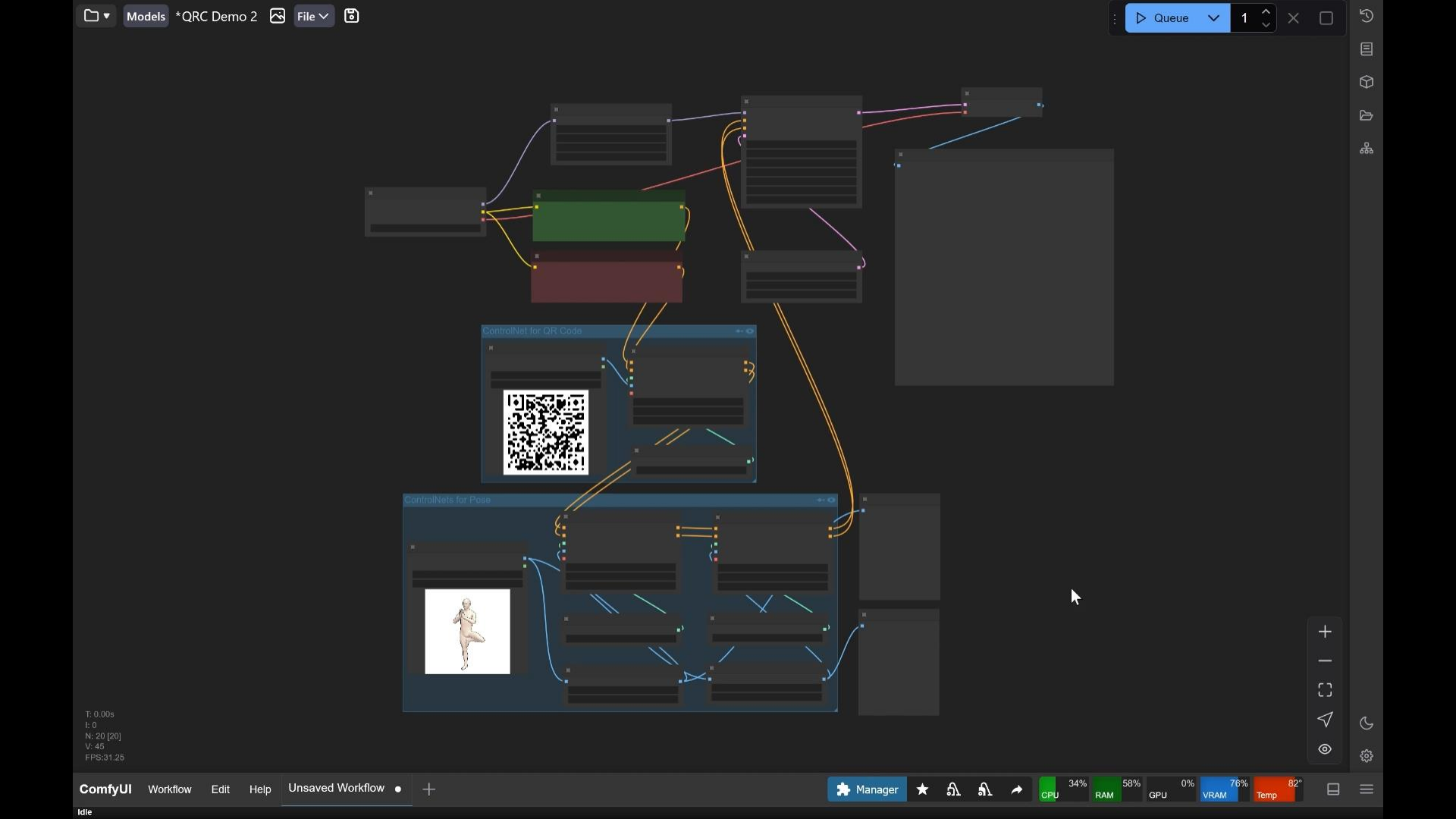
Image en entrée



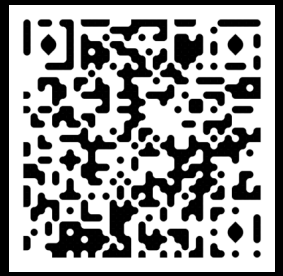
Depth



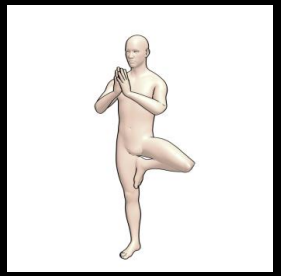
Canny Edge


Models
*QRC Demo 2
File
Queue
1
ControlNet for QR Code
ControlNets for Pose
T: 0.00s
I: 0
N: 20 [20]
V: 45
FPS:31.25
ComfyUI
Workflow
Edit
Help
Unsaved Workflow
Manager
CPU 34%
RAM 58%
GPU 0%
VRAM 76%
Temp 82°
Idle

Générer une image à l’intérieur du QR Code



+

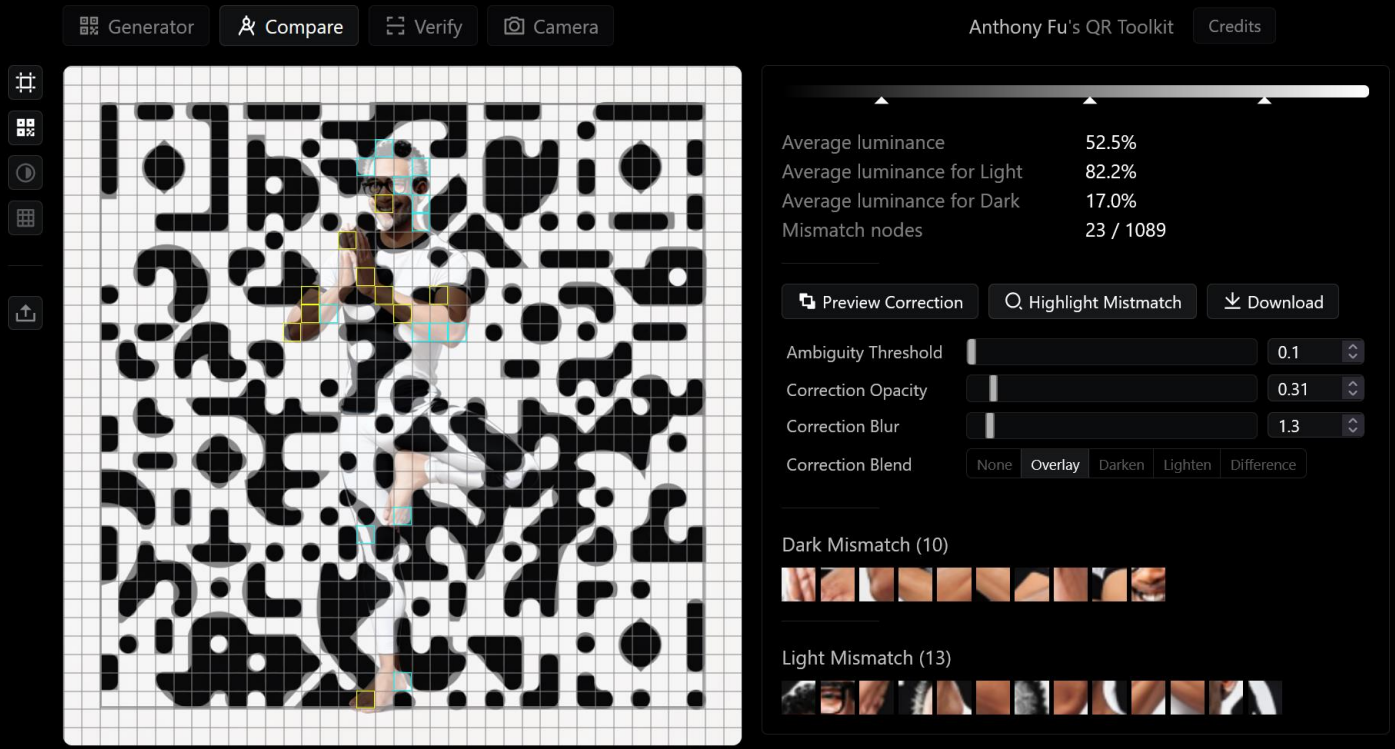


+

1 man, mixed race, short curly hair, black hair, 40 years old, white t-shirt, black yoga pants, short sleeves, smiling, viewing glasses, white background, barefoot



Générer une image à l’intérieur du QR Code

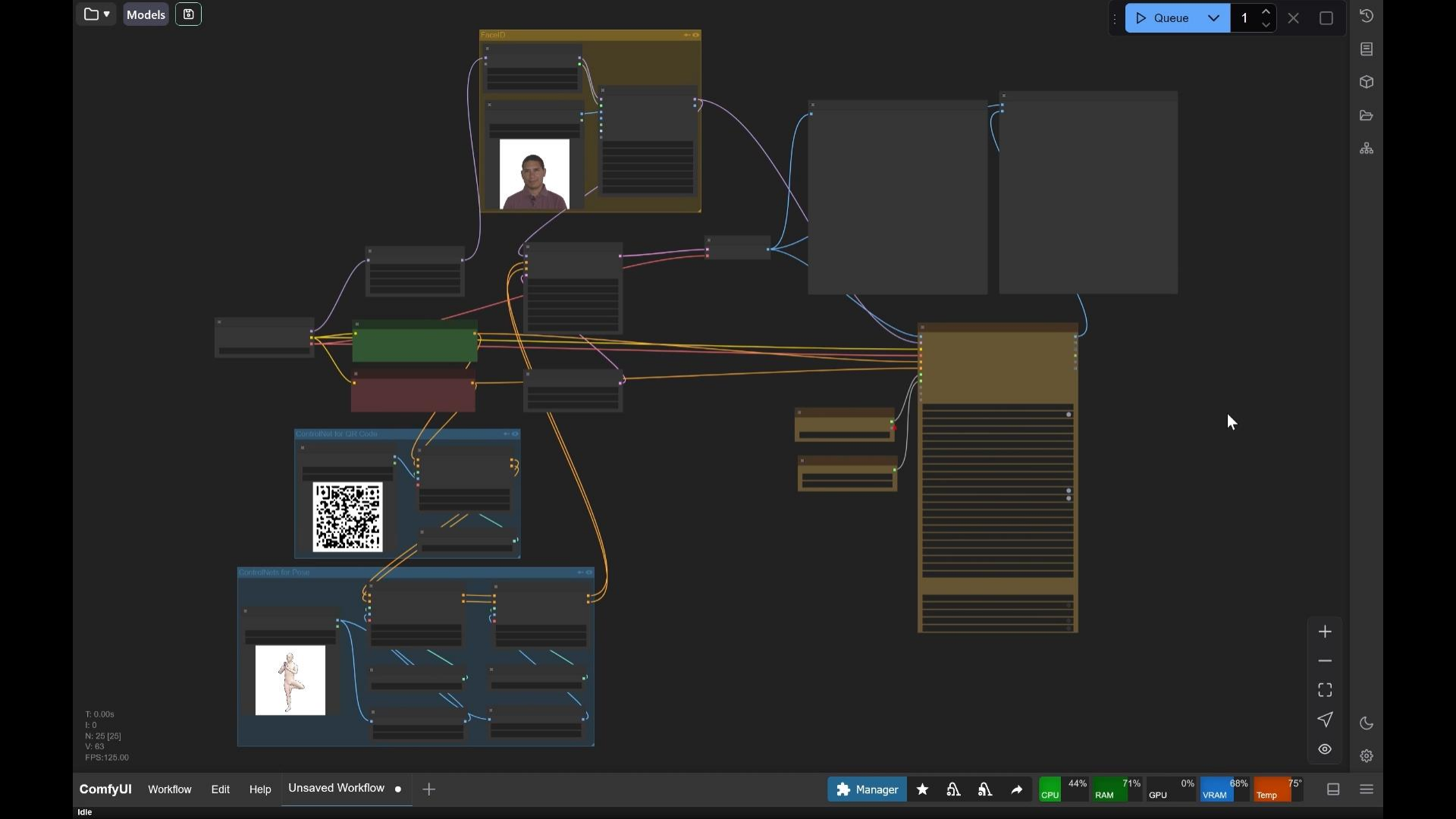


Transférer le visage

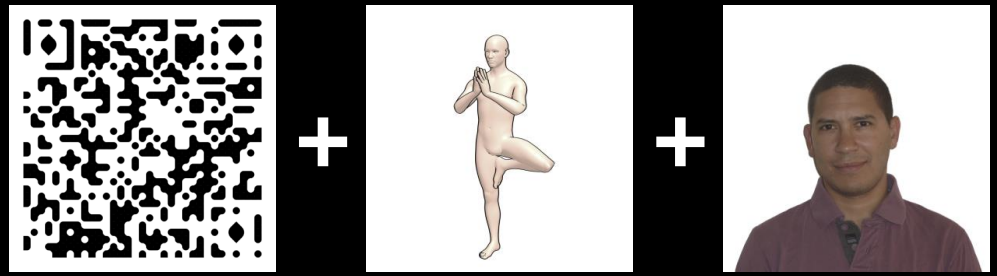
- **Utilisons un *Image Prompt Adapter***
  - Permet d'utiliser une image comme prompt
  - Modèles pré entraînés supplémentaires
  - FaceID IPAdapter : modèle de reconnaissance faciale

- **Et aussi FaceDetailer**
  - Modèle de traitement d'image post génération
  - Amélioration du visage (yeux, nez, lèvres, expression…)




Models
Queue
1
FaceID
ControlNet for QR Code
ControlNets for Pose
T: 0.00s
I: 0
N: 25 [25]
V: 63
FPS:125.00
ComfyUI
Workflow
Edit
Help
Unsaved Workflow
Manager
CPU 44%
RAM 71%
GPU 0%
VRAM 68%
Temp 75°
Idle

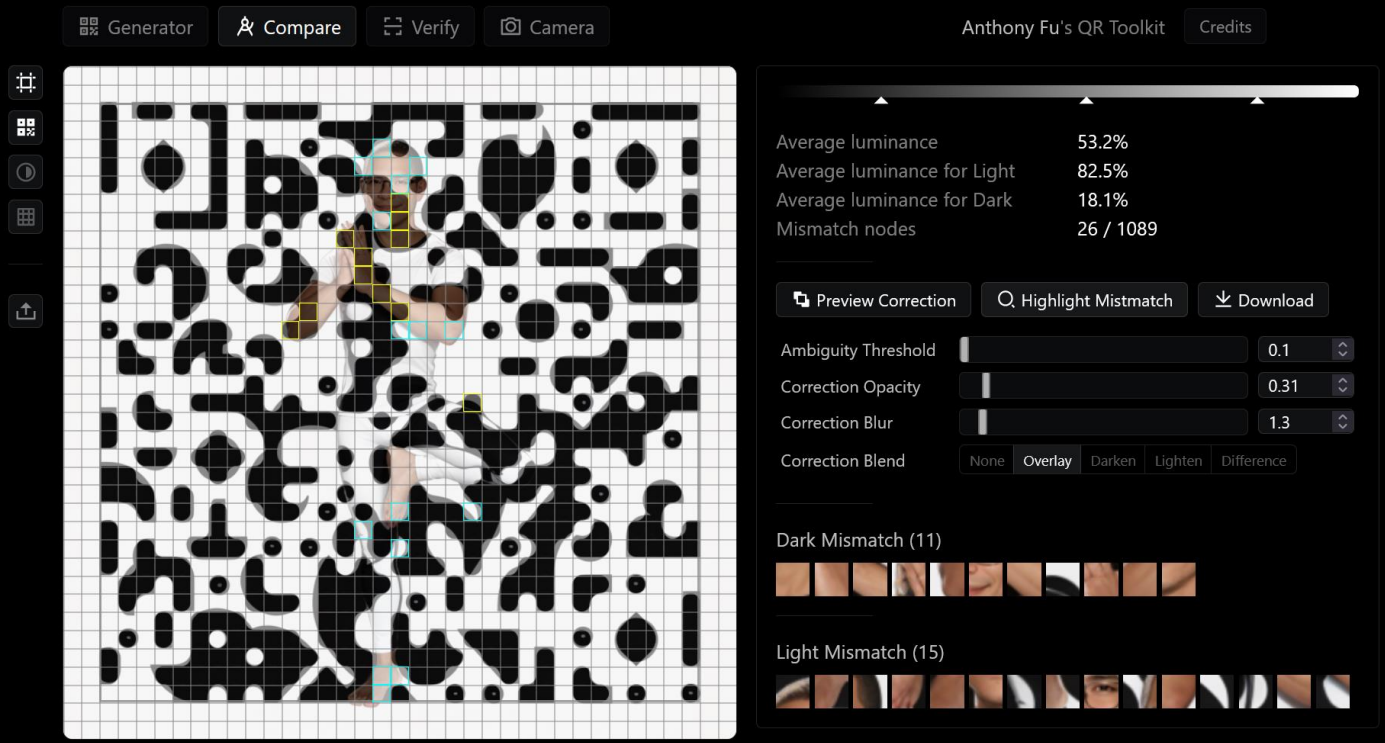
Transférer le visage



1 man, mixed race, short curly hair, black hair, 40 years old, white t-shirt, black yoga pants, short sleeves, smiling, viewing glasses, white background, barefoot



Transférer le visage



Combiner transferts de style, de posture et de visage



Craft | Gobelin Tapestry



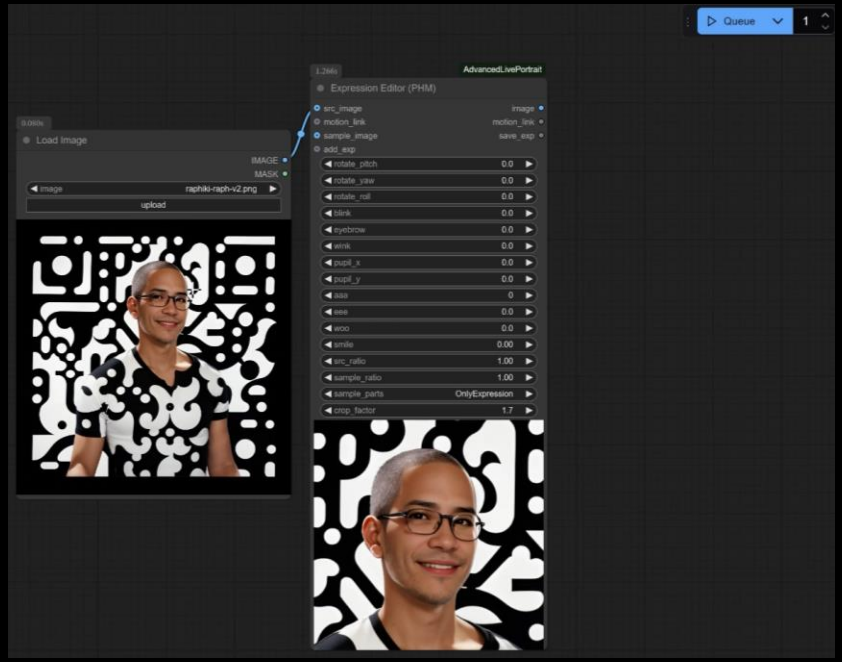
Painting | Watercolor > Cartoon



3D | Abstract > Fluid Dynamics

Comment animer le visage ?

- **Advanced Live Portrait**
  - Modifie et insère des expressions faciales
  - Animations à l’aide de plusieurs expressions
  - Extrait les expressions faciales de photos



Models

Queue 1

AdvancedLivePortrait

AdvancedLivePortrait

VideoHelperSuite

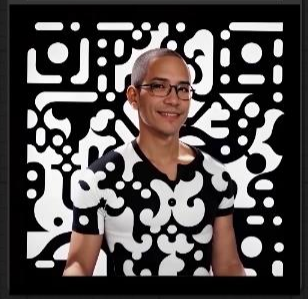
Load Image

IMAGE

MASK

image raphiki-raph-v2....

upload



Expression Editor (PHM)

src_image

motion_link

sample_image

add_exp

image

motion_link

save_exp

rotate_pitch 4.0

rotate_yaw 0.0

rotate_roll 2.0

blink 4.0

eyebrow 6.0

wink 0.0

pupil_x 0.0

pupil_y 0.0

aaa 5

eee 0.0

woo 0.0

smile 1.30

src_ratio 1.00

sample_ratio 1.00

sample_parts OnlyExp...

crop_factor 1.7

Advanced Live Portrait (PHM)

src_images

motion_link

driving_images

images

retargeting_eyes 0.00

retargeting_mouth 0.00

crop_factor 1.7

turn_on true

tracking_src_vid false

animate_without_vid true

```
1 = 5:5
0 = 3:5
```

Video Combine

images

audio

meta_batch

vae

Filenames

frame_rate 15

loop_count 0

filename_prefix AdvancedLivePortrait

format image/gif

pingpong false

save_output true

Note

orginal src : motion 0

Note

motion 1

Note

```
Command Description
1 = 5:5
0 = 3:5

[Motion index] = [Changing frame length] : [Length of frames waiting for next motion]

for example, first line -> 1 = 5:5
Change over 5 frames with motion 1, and wait for the next motion for 5 frames
```

T: 0.00s
I: 0
N: 7 [7]
V: 14
FPS:23.26

ComfyUI Workflow Edit Help Unsaved Workflow (3) Unsaved Workflow (4)

Manager

CPU 29% RAM 59% GPU 0% VRAM 39% Temp 76°

Idle



Merci !

COTRECS

WORLDLINE